Data Lake Storage Backup and Recovery

Protecting the Databricks storage accounts against mistakes, deletions, zone loss and region loss

Prepared for	Baytex: Infrastructure and BI teams
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1. Summary

This document explains how the storage behind Baytex's Databricks platform is protected, how to recover when something goes wrong inside it, such as a pipeline writing bad data or someone deleting files, and how backup differs from disaster recovery. Regional disaster recovery for the whole platform is covered in the companion document, *Disaster Recovery for the Azure Databricks Platform*.

Key points

- **A zone failure loses nothing.** The data lake is zone-redundant (ZRS): every write is stored in three availability zones before it is acknowledged, and Azure switches zones without anyone acting.
- **Most mistakes can already be undone.** Deleted files, folders and containers can be restored for 30 days, Delta tables can be rolled back to an earlier version, and dropped tables can be restored for 7 days.
- **Four risks are not covered today:** nothing stops the storage account itself being deleted; an overwritten file that is not part of a Delta table cannot be recovered, because Azure does not support versioning on an account with a hierarchical namespace; there is no copy outside the account, so a compromised administrator or ransomware could damage data beyond what soft delete covers; and nothing is copied outside Canada Central.
- **Backup and DR solve different problems.** Replication copies mistakes to every copy within minutes. Only a backup, or version history, lets you go back to before the mistake (section 2).
- **The redundancy types stay as they are**: ZRS for the data lake and the workspace root (default) storage, GRS for the Terraform state (section 9).

Recommendations at a glance

#	Recommendation	Closes	Cost (CAD)
R1	Delete lock on each data lake and state account	Account deletion	Free
R2	Alerts on account deletion, lock removal and data-protection changes	Silent loss of protections	Free
R3	30 days of Delta history on critical tables	Bad runs found after 7 days	Extra storage only
R4	Least privilege on the data lake	Bulk deletes by over-privileged identities	Free
R5	Azure Backup vaulted backup for the prod data lake, piloted in dev first	Overwrites, account loss, ransomware, long retention	About 88 per TB a month
R6	Restore drill twice a year	Untested procedures	Free

2. Backup, disaster recovery and high availability

The three are often confused because all of them involve copies of data. They protect against different failures, and none of them replaces the others.

	High availability	Disaster recovery	Backup
Protects against	Loss of a zone: a datacenter group, its power or network	Loss of the whole region	Mistakes and attacks: bad pipeline runs, accidental or malicious deletes, overwrites, ransomware
How it works	Synchronous copies in three zones (ZRS)	A copy or standby in another region (GZRS, managed DR, warm standby)	Point-in-time copies kept for a set period, separate from the live data
Copies mistakes?	Yes, instantly	Yes, within minutes	No: you restore a point from before the mistake

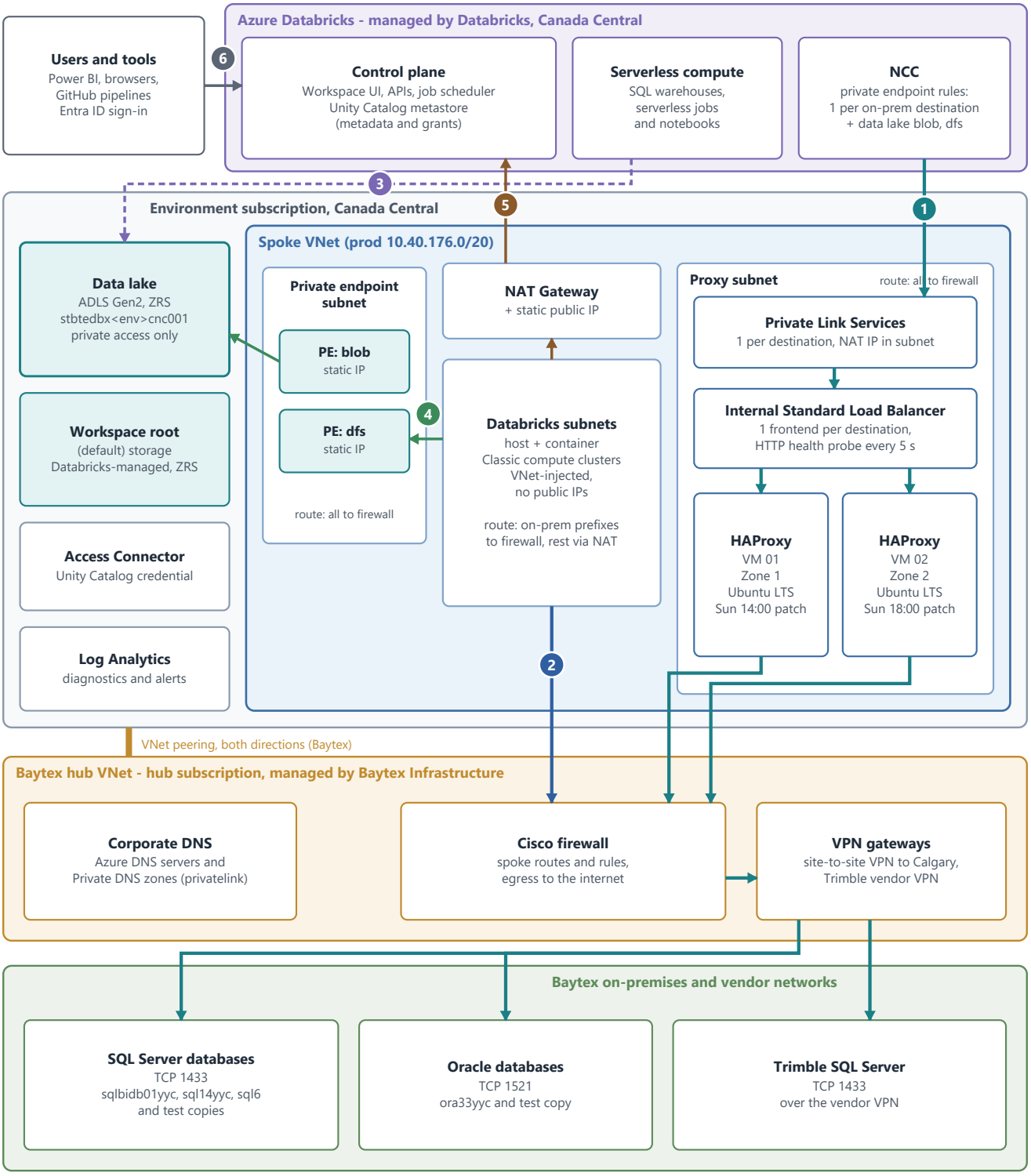
	High availability	Disaster recovery	Backup
Recovery	Automatic, nobody acts	A failover started by people, minutes to days	Restore the affected tables, folders or containers, minutes to hours
Measured by	Uptime	RPO and RTO for the whole service	Retention period and restore-point frequency
Example	Zone 2 in Canada Central fails; nothing visible happens	Canada Central is down; BI runs from Canada East	A pipeline overwrote a table on Tuesday; it is restored to Monday night
In place today	Yes: ZRS	No: see the companion document	Partly: soft delete, Delta history; no copy outside the account

The rule of thumb

Replication, whether across zones or across regions, keeps the data available, and it copies a bad write or a delete to every copy almost at once. Only something that keeps history, such as soft delete, Delta versions or a backup, lets you go back to before the mistake.

3. Current architecture

Every environment (dev, test and prod) is a self-contained spoke in its own Azure subscription in Canada Central, connected to the existing Baytex hub. The diagram shows one environment; prod is used for the example address range. Numbered arrows are the traffic flows described in the table that follows.



Traffic flows

#	From and to	Path
1	Serverless compute to on-premises databases	NCC private endpoint for the destination, then its Private Link Service, the internal load balancer frontend, either HAProxy VM (zone 1 or zone 2), the Cisco firewall and the VPN. HAProxy looks up each database name in corporate DNS when it connects, so a changed database address needs no platform change.
2	Classic compute to on-premises databases	Databricks subnets, whose route table sends the on-premises prefixes to the Cisco firewall, then the VPN.
3	Serverless compute to the data lake	NCC private endpoints on the data lake's blob and dfs endpoints.
4	Classic compute to the data lake	The blob and dfs private endpoints in the spoke, at fixed addresses.
5	Classic compute to the internet and the Databricks control plane	NAT Gateway with one static public IP. Everything else from the Databricks subnets that is not an on-premises prefix leaves this way.
6	Users, Power BI and pipelines to the workspace	Public workspace URL, signed in with Microsoft Entra ID. GitHub pipelines use short-lived OIDC tokens.

How the HAProxy tier keeps on-premises access available

- Two Ubuntu VMs run HAProxy, one in availability zone 1 and one in zone 2, behind an internal Standard Load Balancer. Each on-premises destination has its own load balancer frontend and its own Private Link Service, which is what the Databricks NCC connects to (seven destinations in prod).
- The load balancer probes each VM every five seconds and removes a VM after one failed probe, so either VM can carry all traffic while the other is restarting, being patched or lost with its zone.
- The VMs are patched by Azure Update Manager on Sundays, one after the other: zone 1 from 14:00 to 16:00 and zone 2 from 18:00 to 20:00 Mountain Time.
- The proxy subnet sends all traffic to the Cisco firewall, so the firewall rules and the VPN decide which on-premises hosts the platform can reach. Baytex adds the proxy subnet to its on-premises return routes.

4. The storage accounts and how they are protected today

Account	Holds	Managed by	Type	Protection today
Data lake, stbtedbx<env>cnc001	Unity Catalog managed and external tables, files and volumes, in the unity-catalog container	Terraform (AMTRA)	ZRS	30-day soft delete for files, folders and containers; private network access only; Entra ID sign-in only (no account keys); availability alert. Versioning and change feed are off by design, because the hierarchical namespace does not support them
Workspace root (default) storage, stbtedbx<env>root001	Workspace system data, the DBFS root and the workspace catalog's default storage	Azure Databricks, in its managed resource group	ZRS	Databricks controls its settings and a deny assignment protects it; not for business data
Terraform state	The state of every resource Terraform manages	Bootstrap pipeline (AMTRA)	GRS	Versioning and 30-day soft delete, so a damaged state file can be rolled back

Why the data lake's options are narrower

The data lake has a hierarchical namespace, which gives it real folders and is what Databricks recommends for a data lake. Azure does not yet offer several blob-storage protections for this kind of account:

Feature	With a hierarchical namespace	Effect for Baytex
Soft delete for files, folders and containers	Supported	On, 30 days
Blob versioning (keeps every overwritten version)	Not supported	Off by design: the Terraform sets `versioning_enabled = false`. An overwritten non-Delta file cannot be recovered from the account itself
Point-in-time restore	Not supported	The account cannot be rolled back as a whole
Change feed and object replication	Not supported	Off by design: the Terraform sets `change_feed_enabled = false`. No do-it-yourself copy to another account
Immutable (write-once) storage	Supported	Available for containers that must never change
Azure Backup vaulted backup for Data Lake Storage	Supported, generally available in all regions	The main way to keep a separate copy (recommendation R5)
Customer-started account failover (GRS or GZRS)	Supported	Covered in the DR document

Versioning and change feed are switched off in the data lake's configuration on purpose, not overlooked. Azure accepts only the off value for an account with a hierarchical namespace, so they cannot be switched on unless Microsoft adds support. The protection they would give comes instead from Delta history, soft delete and, if adopted, vaulted backup.

Most data in the lake is in Delta tables managed by Unity Catalog. Delta keeps its own history: every change writes new files and records a new table version, and old files are removed only by VACUUM after the retention period, 7 days by default. This is the first line of defence against bad pipeline runs.

5. What can go wrong, and how to recover

Scenario	Protected today?	How to recover	Data that can be lost	Recommended addition
1. A pipeline writes wrong data into a Delta table: a bad MERGE, UPDATE, DELETE or overwrite	Yes: Delta history, about 7 days by default	Roll the table back (procedure A)	Changes made after the point you restore to	R3: keep 30 days of history on critical tables
2. A table is dropped	Managed tables: yes, for 7 days. External tables: the files stay in storage	UNDROP the managed table (procedure B); recreate an external table over its folder	None within 7 days	R4: limit who can drop tables
3. Files or folders are deleted directly in storage, by a job or a person	Yes: soft delete, 30 days	Restore the deleted paths (procedure C)	None within 30 days	-
4. A container is deleted	Yes: container soft delete, 30 days	Restore the container (procedure D)	None within 30 days	-
5. A file outside a Delta table is overwritten	No: versioning is not supported with a hierarchical namespace, so it is off by design	Only from a backup	The previous content	R5, or land raw files once and never rewrite them
6. The storage account is deleted	No	Microsoft can sometimes recover it within 14 days (procedure F); not guaranteed	Possibly everything	R1 delete lock; R5 backup
7. A compromised administrator or ransomware deletes or encrypts data	Partly: deleted data stays in soft delete for 30 days; encrypted overwrites cannot be undone	Soft delete for deletes; a backup for overwrites	Everything overwritten	R2, R4, R5
8. Soft delete is switched off or shortened by mistake	No alert today	The next deploy restores the setting from Terraform	Anything deleted while it was off	R2 alert
9. One availability zone fails	Yes: ZRS	Automatic; nothing to do	None	-
10. Canada Central is lost	No: every copy is in Canada Central	Wait for the region, or fail over (DR document)	None while the region exists; the data is unavailable meanwhile	GZRS or managed DR (DR document)
11. A network or access change blocks the account	Yes: settings are in Terraform	Redeploy, which restores the settings	None; access only	Keep changes in the pipeline

A note on deleted Delta files

When VACUUM removes old table files, soft delete keeps them for 30 days, and they are billed as normal storage while they wait. In an emergency this means a table can sometimes be recovered past its Delta retention by restoring the files and then rolling the table back. It is a last resort, worked through with AMTRA and Baytex BI, not a routine procedure.

6. Recovery procedures

Before you start

- The data lake accepts connections only through its private endpoints. Run the Azure commands from a machine on the Baytex network that resolves the storage names to their private addresses; the Azure portal's data views need the same path.
- Restoring files needs the Storage Blob Data Contributor role on the account. Restoring tables needs the matching Unity Catalog privileges, which Baytex BI holds.
- Stop the pipeline that caused the problem first, so it does not repeat the damage after the restore.

A. Roll a Delta table back to before a bad run

Owner: Baytex BI. Run in a Databricks SQL editor or notebook.

```
-- 1. Find the last good version: each write is listed with its time and operation
DESCRIBE HISTORY bte_prod.sales.orders;

-- 2. Optional: check the data at that version before restoring
SELECT COUNT(*) FROM bte_prod.sales.orders VERSION AS OF 412;

-- 3. Restore to that version, or to a time
RESTORE TABLE bte_prod.sales.orders TO VERSION AS OF 412;
RESTORE TABLE bte_prod.sales.orders TO TIMESTAMP AS OF '2026-10-06T23:00:00';
```

A restore is itself a new version, so it can be undone. If it fails because the files were already removed by VACUUM, see the note at the end of section 5. The catalog, schema and table names above are examples.

B. Restore a dropped managed table

Owner: Baytex BI. Works for 7 days after the drop.

```
SHOW TABLES DROPPED IN bte_prod.sales;
UNDROP TABLE bte_prod.sales.orders;
```

C. Restore deleted files or folders

Owner: AMTRA or Baytex Infrastructure. Works for 30 days after the delete. A folder is restored together with everything that was in it. If something new now exists at the same path, move it aside first.

```
Connect-AzAccount -Tenant <tenant-id>
$ctx = New-AzStorageContext -StorageAccountName "stbtedbxprodcnc001" -UseConnectedAccount

# List what was deleted under a folder of the unity-catalog container
$deleted = Get-AzDataLakeGen2DeletedItem -Context $ctx -FileSystem "unity-catalog" -Path "<folder>"
$deleted | Format-Table

# Restore it
$deleted | Restore-AzDataLakeGen2DeletedItem
```

D. Restore a deleted container

Owner: AMTRA or Baytex Infrastructure. Works for 30 days. In the portal: the storage account, Containers, Show deleted containers, then Undelete. With PowerShell:

```
Get-AzStorageContainer -Context $ctx -IncludeDeleted | Where-Object IsDeleted
Restore-AzStorageContainer -Context $ctx -Name "unity-catalog" -VersionId "<version-id>"
```

E. Restore from a vaulted backup (once R5 is in place)

1. In the Backup vault, choose the recovery point from before the problem and the containers, or folder prefixes, to restore.
2. Restore to a separate storage account in Canada Central. Azure Backup does not restore into the account that was backed up, and the target must not already have containers with the same names.
3. Check the restored data, then copy what is needed back into the data lake, or have Baytex BI point the affected tables at the restored location.

F. A deleted storage account

1. Act at once. In the Azure portal, open Storage accounts and use the option to restore a deleted account. It is available for up to 14 days if the resource group still exists and no new account has taken the name. If the account is not offered, open a support case with Microsoft immediately.
2. Run a deploy for the environment, so Terraform restores the private endpoints, settings and role assignments around the account.
3. Check Unity Catalog access with a test query through the external location.

Recovery of a deleted account is best effort. The delete lock (R1) is what prevents the situation.

7. Recommendations

R1. Delete lock on each data lake and state account

A CanNotDelete lock stops anyone, including a pipeline or an administrator in a hurry, from deleting the account, while every other change keeps working. Removing the lock needs Owner or User Access Administrator, so the deployment identity, which has Contributor, can neither remove it nor create it: a subscription Owner at Baytex creates it once. The Destroy workflow then stops at the locked account by design. The documented teardown adds a step for an Owner to remove the lock first. Cost: free.

R2. Alerts that catch protections being removed

Activity log alerts, which are free, on: deletion of a storage account; deletion of a lock; and changes to the data lake's blob service settings, which is where soft delete is set. Optionally, a metric alert on an unusually high number of delete operations. AMTRA adds these to the Terraform alongside the existing alerts.

R3. 30 days of Delta history on critical tables

Owner: Baytex BI. A bad run that is noticed after a week can no longer be rolled back with the 7-day default. Keeping 30 days on the tables that matter costs only the storage of the extra file versions, which depends on how much of each table changes:

```
ALTER TABLE bte_prod.sales.orders SET TBLPROPERTIES (
  'delta.deletedFileRetentionDuration' = 'interval 30 days',
  'delta.logRetentionDuration'         = 'interval 30 days');
```

R4. Least privilege on the data lake

- Review who holds Storage Blob Data Contributor, Storage Account Contributor or Owner on the data lake. People and jobs reach the data through Unity Catalog, which needs none of them.
- The data Access Connector holds three extra roles only for Auto Loader file events, which are switched off on the external location. While they stay off, those roles can be removed, which also stops the connector from changing the account's settings.
- Baytex BI grants table-level privileges such as MODIFY and the right to drop tables to the job identities and groups that need them, not to everyone.

R5. Azure Backup vaulted backup for the prod data lake

Vaulted backup keeps a copy of chosen containers in a Backup vault, outside the storage account, once a day, for up to ten years. It covers what nothing else does: overwritten files, a deleted account, and an attacker who controls the account. Recommended set-up: daily backups kept 30 days, adding monthly backups kept 12 months if Baytex needs long-term retention; a ZRS vault with soft delete and immutability switched on, so recovery points cannot be removed early.

Trade-offs to accept before switching it on

- **Undelete stops working while backup is on.** Files are still soft-deleted, but Azure does not allow them to be restored from soft delete; deleted files then come from the last daily backup, which can lose up to a day of changes. This is the main reason to pilot it in dev first.
- **Restores go to another account** in the same region, and data is then copied back (procedure E). Vaulted backup is therefore not a regional DR solution.
- **One scheduled backup a day**, plus up to four on demand.
- **The data lake's Terraform settings stay as they are.** Azure Backup copies the data with its own platform replication, which does not need versioning or change feed, so both stay off. The dev pilot confirms that a deploy leaves the backup untouched.
- **The vault needs the Storage Account Backup Contributor role** on the account, and the account must not be associated with a network security perimeter (it is not today).

R6. Restore drill twice a year

In dev: roll back a table, undrop a table, restore a deleted folder and a deleted container, and, once R5 is in place, restore from the vault. Record how long each took. Owners: AMTRA and Baytex BI.

Optional: write-once landing data

If raw source extracts must be kept unchanged for audit, an immutable (write-once) policy on a dedicated container makes them impossible to change or delete for the retention period, even by an administrator. Use it only where that is a requirement, because it also blocks legitimate clean-up.

8. Cost

Azure list prices in Canadian dollars (pay-as-you-go) for Canada Central, from the Azure Retail Prices API on 7 October 2026. Taxes and Baytex's discount agreement are not included.

Item	Price	Example
Soft delete (on today)	Deleted data is billed as normal storage until it expires: ZRS 0.0354 per GB a month	Tables that rewrite 10% of their data a week keep roughly 40% more data in soft delete
Delete lock (R1), activity log alerts (R2)	Free	-
Longer Delta history (R3)	Storage of the extra file versions	Depends on how much each table changes
Vaulted backup (R5): protected instance	17.00 a month per account of 1 TB or more; 60% for 100 GB to 1 TB, 30% for 10 to 100 GB, 10% under 10 GB	-
Vaulted backup (R5): vault storage	ZRS 0.0436, LRS 0.0348, GRS 0.0748 per GB a month	-
Vaulted backup, 1 TB, 30 daily points, 2% daily change, ZRS vault	About 1,638 GB stored	About CAD 88 a month
Vaulted backup, 5 TB, same policy, ZRS vault	About 8,192 GB stored	About CAD 374 a month
Vaulted backup, 1 TB, same policy, GRS vault	About 1,638 GB stored	About CAD 140 a month

Item	Price	Example
Data lake ZRS to GZRS (regional copy, DR document)	+28.98 per TB a month, plus 0.0283 per GB written	2 TB: about CAD 64 a month

Vault storage depends on how much data changes each day and how long recovery points are kept; the examples assume 2% of the data changes daily. Backup write operations add a small amount on top.

9. Storage redundancy recommendation

Each environment has two storage accounts that Databricks depends on: the **data lake** (stbtedbx<env>cnc001), which holds Unity Catalog tables and files, and the **workspace root (default) storage** (stbtedbx<env>root001), which Azure Databricks creates and manages for the workspace's own system data. A third account holds the Terraform state. The redundancy type decides which failures a storage account survives without anyone acting.

The options

Type	Copies of the data	Survives a zone loss	Survives a region loss	CAD per TB-month (hot)
LRS	3 copies in one datacenter	No	No	28.98
ZRS	3 copies, one in each of three zones	Yes, with no action	No	36.25
GRS	3 local copies + 3 in Canada East, copied asynchronously	No: the primary copy is single-zone	Yes, after a failover you start	58.06
GZRS	3 zonal copies + 3 in Canada East, copied asynchronously	Yes, with no action	Yes, after a failover you start	65.23
RA-GRS, RA-GZRS	As GRS or GZRS, plus a read-only endpoint in Canada East	As GRS or GZRS	As GRS or GZRS, and readable before failover	72.50, 81.61

Prices are list prices for Data Lake Storage (hierarchical namespace) hot data in Canada Central. Geo-copies trail the primary, typically by less than 15 minutes, although Microsoft gives no guarantee. Canada East has no availability zones, so ZRS and GZRS cannot be chosen for an account that lives there.

Recommendation

Storage account	Today	Recommended	Why
Data lake, dev and test	ZRS	**ZRS** (no change)	Keeps dev and test behaving like prod. LRS would save about CAD 7 per TB a month but would remove zone protection from the environments used to prove changes.
Data lake, prod	ZRS	**ZRS** now. Move to **GZRS** only if Baytex chooses regional DR option A (rebuild in Canada East)	ZRS already survives the loss of a zone with no data loss. GZRS adds an asynchronous copy in Canada East for about CAD 28.98 more per TB a month. With managed DR (option B) or a warm standby (option C) the copy is made by Databricks or by your own jobs, so the primary stays ZRS. GRS is not recommended because it gives up zone protection, and the read-access variants add cost without helping Databricks, whose table paths always use the primary endpoint.
Workspace root (default) storage, all environments	ZRS	**ZRS** (no change)	A workspace cannot fail over to another region, and a new workspace cannot reuse another workspace's root storage, so a geo-copy adds cost with little recovery value. ZRS gives zone protection. Azure's default for this account is GRS; the platform sets ZRS deliberately. Business data belongs in the data lake, not here.
Terraform state, all environments	GRS with versioning and 30-day soft delete	**GRS** (no change)	The copy in Canada East lets the pipelines deploy a recovery environment after a Canada Central outage. The account is tiny, so cost is negligible.
Backup vault, if vaulted backup is adopted	Not used	**ZRS** vault	Recovery points survive a zone loss. Restores go to an account in the same region, so a GRS vault adds little unless Baytex also wants the backups copied out of Canada Central.
Standby accounts in Canada East (DR options B and C)	Not used	**LRS**	Canada East offers no zones. GRS would copy the standby back to Canada Central, which the primary already covers.

Changing the prod data lake from ZRS to GZRS

Adding geo-redundancy to a ZRS account is an online change made by Azure. It is made through Terraform by changing the data lake's replication type and running a deploy. Before approving, confirm the plan shows the storage account being **updated in place**. A plan that says the account **must be replaced** would delete the data lake and must be rejected. A delete lock on the account (recommendation R1 of the storage backup document) is a second safeguard against that mistake.

10. Next steps

Step	Owner
Create the delete locks on the data lake and state accounts (R1), and add the teardown step to the runbook	Baytex Infrastructure (Owner), AMTRA
Add the alerts (R2) and review data lake access (R4) through the pipeline	AMTRA
Set 30-day Delta history on critical tables (R3); review drop and modify grants	Baytex BI
Decide on vaulted backup (R5); pilot it in dev, including a full restore	Baytex, AMTRA
Run the first restore drill (R6) in dev	AMTRA, Baytex BI

11. Sources

Microsoft documentation, read on 7 October 2026.

- Blob Storage feature support in storage accounts: learn.microsoft.com/azure/storage/blobs/storage-feature-support-in-storage-accounts
- Support matrix for Azure Data Lake Storage vaulted backup: learn.microsoft.com/azure/backup/azure-data-lake-storage-backup-support-matrix
- Configure vaulted backup for Azure Data Lake Storage: learn.microsoft.com/azure/backup/azure-data-lake-storage-configure-backup
- Configure and manage backup for Azure Blobs: learn.microsoft.com/azure/backup/blob-backup-configure-manage
- Reliability in Azure Databricks: learn.microsoft.com/azure/reliability/reliability-databricks
- Azure Retail Prices API (list prices, CAD): prices.azure.com/api/retail/prices